Q1.1:

- Content creators and accessibility staff at public entities need to ensure that covered Excel workbooks are accessible to people with disabilities and remain ready for the applicable compliance deadline.
- Current approaches rely on Microsoft's built-in Accessibility Checker, manual review, and separate tracking methods, which require specialized knowledge and do not provide a centralized way to remediate issues, monitor workbook versions, or preserve compliance evidence.
- Software could improve this process by allowing users to upload workbooks for accessibility evaluation, automatically repairing supported issues, guiding users through problems requiring human judgment, and periodically auditing stored workbooks for changes or unresolved barriers.
- This problem relates to the requirement that covered web content and mobile applications conform to WCAG 2.1 Level A and Level AA under **28 C.F.R. § 35.200**, including Excel workbooks that qualify as "conventional electronic documents" under **28 C.F.R. § 35.104**. [1][2]

Q1.2:

- Phase 1 explored several general directions, including accessibility checking and remediation across multiple document formats, document inventory and monitoring, and tracking evidence for regulatory compliance or exceptions.
- In addition, we felt that simple checking software is overdone and we wanted to be unique.
- After comparing the teams' broader problem areas, the pod selected an Excel accessibility hub because it combines automated accessibility evaluation and remediation with centralized monitoring and compliance evidence while maintaining a focused scope.
- The problem involves meaningful engineering work because the system must inspect and modify complex workbooks without damaging their data, formulas, formatting, references, or functionality.
- Focusing on .xlsx workbooks makes the project achievable within the semester while still allowing the team to implement detection, guided or automatic remediation, reporting, and periodic auditing.

- Independent implementations can be evaluated using the same test workbooks and compared based on issue-detection accuracy, successful remediations, unresolved issues, and preservation of workbook functionality.

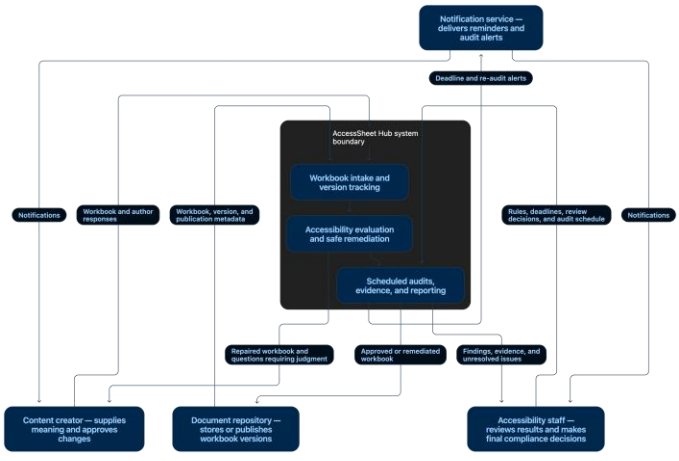
Q1.3:
Meeting: 9/20/26, 1 pm - ,
Team 9
Jackson Graff
Dylan Patel
Dixon Wagner

Ryan Covitz
Ryan
Team 12
Jonathan Shi
Gustavo Santana
Haruki Kuwasawa
Logan Hanson

Set this version as version 1.0

No disagreements remain

Q2.1:



Q2.2:

Included capabilities

- Accept a copy of an .xlsx workbook and evaluate it for a shared set of accessibility issues connected to WCAG 2.1 Level A and Level AA under **28 C.F.R. § 35.200**.
- Display detected issues with their worksheet and cell or object locations, explanations, and recommended corrections.
- Apply supported remediations to a **new copy** of the workbook without changing or overwriting the original uploaded file.
- Allow the user to download the remediated workbook and a version-specific audit report documenting findings, changes, and unresolved issues.
- Retain audit records and periodically remind users to upload or submit updated versions for reassessment before applicable compliance deadlines.

Explicitly excluded or limited capabilities

- The system will not modify, overwrite, or automatically publish the original or live workbook.
- Periodic audits will evaluate the version stored in the hub; they cannot detect changes to an external live workbook unless a new version is uploaded or a repository integration is added.
- The system will not make final legal-compliance or regulatory-exception decisions.
- Meaning-dependent changes, such as writing alternative text or reorganizing complex tables, will require author or reviewer input.
- Formats other than .xlsx, along with encrypted, corrupted, password-protected, or macro-enabled workbooks, are outside the required common scope.

Q2.3

Team 9 estimate

- **Design and requirements: 80–105 person-hours**
- **Implementation: 65–90 person-hours**
- **Integration: 30–45 person-hours**
- **Validation: 50–70 person-hours**
- **Deployment: 15–25 person-hours**
- **Estimated total: 240–335 person-hours over 10 weeks**, or approximately **24–34 team person-hours per week**.
- **GenAI assumptions:** Team 9 has access to limited general-purpose GenAI tools, mainly just the free student plan for GPT. GenAI is going to assist with code examples, test-case generation, documentation, and debugging, and reducing some implementation effort. It

will not substantially reduce design or validation time because generated recommendations must be independently verified.

Team 12 estimate

- **Design and requirements: 80–105 person-hours**
- **Implementation: 65–90 person-hours**
- **Integration: 30–45 person-hours**
- **Validation: 50–70 person-hours**
- **Deployment: 25–40 person-hours**
- **Estimated total: 250–350 person-hours over 10 weeks**, or approximately **25–35 team person-hours per week**.
- **GenAI assumptions:** Team 12 has access to limited general-purpose GenAI tools, mainly just the free student plan for GPT. GenAI is going to assist with code examples, test-case generation, documentation, and debugging, and reducing some implementation effort. It will not substantially reduce design or validation time because generated recommendations must be independently verified.
- **Reason for the difference:** The teams have comparable design, implementation, integration, and validation estimates because they are delivering the same scope. Team 12's total is slightly higher because it expects to spend an additional **10–15 person-hours** learning and troubleshooting deployment.

Q3.1

- **REQ-01 — Supported workbook intake:** When an authenticated user uploads a valid .xlsx workbook that satisfies the supported limits defined and justified through prototype performance testing in Section 5, the system shall accept it and create an audit record.
  Basis: Establishes a measurable scope that both teams can implement and test within the semester.
- **REQ-02 — Invalid or unsupported input:** When a user uploads a file that is not a valid supported .xlsx workbook, the system shall reject it without beginning an audit and shall explain the reason. Unsupported inputs include .xls, .xlsm, CSV, Google Sheets links, password-protected files, encrypted files, and corrupted files.
  Basis: Prevents unsafe processing and avoids presenting incomplete results as valid accessibility audits.
- **REQ-03 — Original workbook protection:** While processing a workbook, the system shall treat the uploaded source as read-only and shall not modify, overwrite, or publish the original or any external live workbook.

Basis: Stakeholders require remediation to produce a separate downloadable copy so that formulas, content, or functionality in the live workbook are not unintentionally changed.

- **REQ-04 — Accessibility evaluation:** For every accepted workbook, the system shall perform the common accessibility checks defined in Section 5. At minimum, the checks shall cover missing alternative text, table-header problems, merged cells in populated regions, generic worksheet names, ambiguous hyperlink text, and potential color or contrast problems.
  Basis: Spreadsheet files are conventional electronic documents under **28 C.F.R. § 35.104**, and covered web content must conform to WCAG 2.1 Level A and Level AA under **28 C.F.R. § 35.200**.
- **REQ-05 — Actionable findings:** When the system identifies an accessibility issue, it shall report the affected worksheet and cell range or object, explain the barrier, recommend a response, and classify the issue as automatically remediable, requiring user input, or requiring manual review.
  Basis: Content creators may not have specialized accessibility knowledge and need enough information to understand and address each finding.
- **REQ-06 — Human judgment:** When a correction depends on the workbook's intended meaning, the system shall request user input or mark the issue for manual review rather than generating the missing information. Meaning-dependent issues include alternative-text content, descriptive worksheet names, logical table organization, and whether color conveys essential information.
  Basis: Automated software cannot reliably determine an author's intended meaning, and an incorrect repair could make the workbook less understandable.
- **REQ-07 — User-authorized remediation:** When a supported remediation is available, the system shall apply it only after the user selects or approves the correction. Supported remediation shall include inserting user-provided alternative text, applying a user-provided worksheet name, improving hyperlink text, and marking a user-confirmed header row where technically supported.
  Basis: Stakeholders need guided remediation without allowing the system to make unapproved semantic changes.
- **REQ-08 — Separate remediated output:** When remediation finishes successfully, the system shall generate a new downloadable .xlsx workbook while retaining the uploaded source as a separate, unchanged version.
  Basis: This provides users with a candidate replacement workbook without modifying the current live document.
- **REQ-09 — Workbook preservation:** For every generated workbook, the system shall preserve all formulas, cell values, worksheet references, named ranges, formatting, and supported features not explicitly selected for remediation. The generated file shall open in Microsoft Excel without a corruption or repair warning.

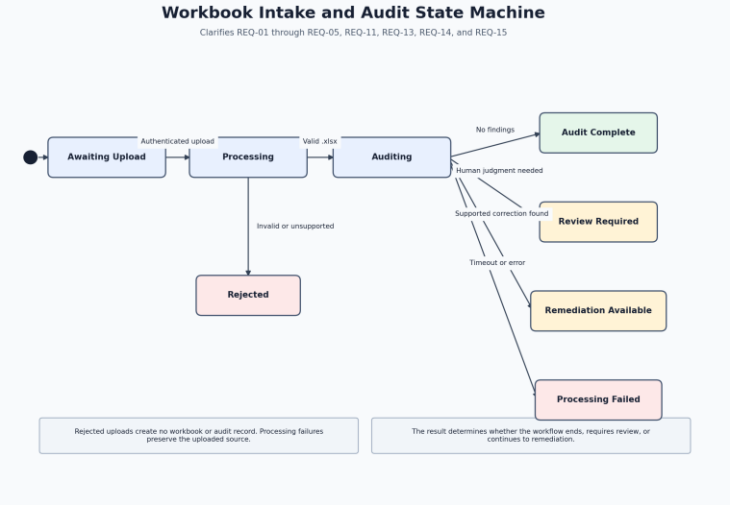
Basis: Accessibility improvements must not damage the workbook's existing content or behavior.

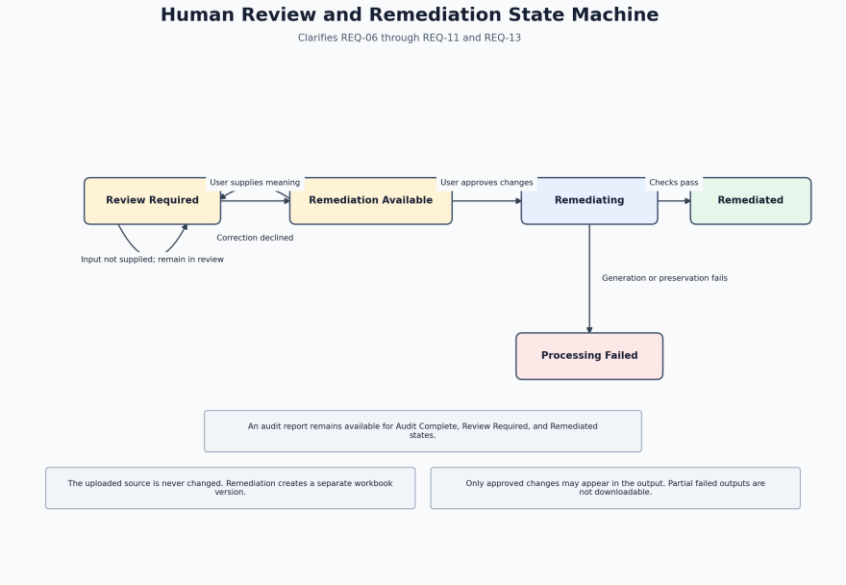
- **REQ-10 — Audit report:** After an audit or remediation completes, the system shall provide a report containing the workbook version, audit date, ruleset version, checks performed, findings, approved changes, unresolved issues, manual-review items, and processing limitations.
  Basis: Accessibility staff need version-specific evidence showing what was checked, changed, and left unresolved.
- **REQ-11 — Version and status tracking:** When a workbook is uploaded, remediated, or re-audited, the system shall create a uniquely identified version and preserve earlier audit records. Each version shall have a status of **Processing**, **Review Required**, **Remediation Available**, **Remediated**, Audit Complete, or **Processing Failed**.
  Basis: Accessibility results apply only to the exact workbook version evaluated. The system shall not use a status of "ADA compliant" because final compliance decisions remain with authorized personnel.
- **REQ-12 — Periodic audits and deadline reminders:** When periodic auditing is enabled, the system shall recheck the stored workbook version at a user-configured interval between **1 and 90 days**. When an administrator provides the applicable compliance deadline, the system shall issue default reminders **90, 30, and 7 days** before that date for workbooks with unresolved findings or no completed audit. Each result shall state that changes to an external workbook cannot be detected until a new version is uploaded.
  Basis: Public entities must continue maintaining accessible content after their applicable deadline, and DOJ guidance recommends regular accessibility testing. The current deadlines are April 26, 2027, or April 26, 2028, depending on the public entity.
- **REQ-13 — Controlled failure handling:** If uploading, auditing, remediation, report generation, or scheduled auditing fails, the system shall preserve the source workbook, record the failure, display an understandable error message, and not present a partial workbook or report as successfully completed.
  Basis: Users must not lose their source file or rely on incomplete accessibility results.
- **REQ-14 — Access control and data protection:** The system shall require authentication and shall prevent users from accessing workbook content or reports they do not own or have permission to review. The deployed system shall use HTTPS, shall not store plaintext passwords, and shall allow an authorized user to delete stored workbook files and generated outputs.
  Basis: Uploaded spreadsheets may contain student, employee, financial, or other sensitive institutional information.
- **REQ-15 — Accessibility, performance, and repeatability:** The user interface shall support keyboard operation, visible focus, programmatic form labels, text-based errors, and WCAG 2.1 AA color contrast. In the documented test environment, the system shall

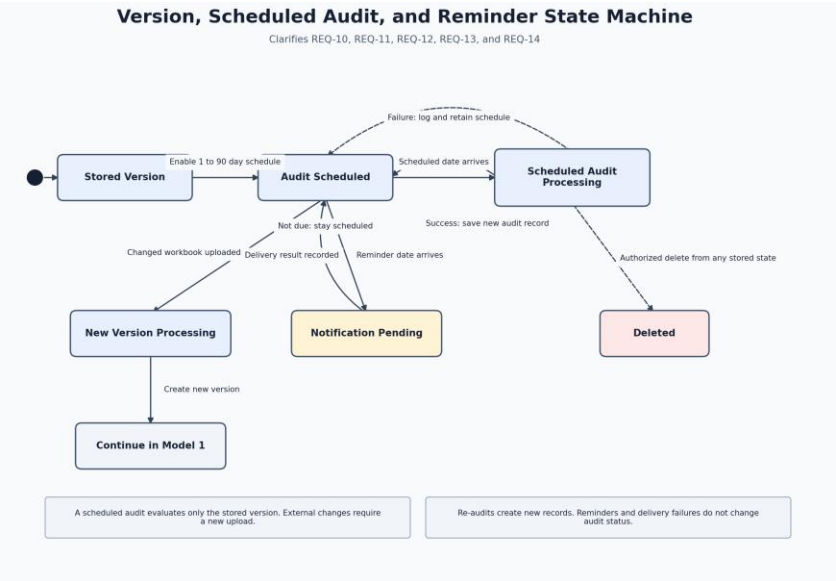
audit a workbook no larger than 25 MB with no more than 250,000 **nonempty cells** within **60 seconds** or return a controlled failure. Repeated audits of the same workbook using the same ruleset version shall produce the same automated findings.
Basis: The hub must not create new accessibility barriers, must provide results within a usable amount of time, and must produce results that can be independently compared between teams. WCAG 2.1 provides testable Level A and AA success criteria.

Categories that do not apply or are excluded

- **Other document formats are excluded.** PDF, Word, PowerPoint, CSV, Google Sheets, .xls, and .xlsm support are outside the common scope because supporting several file structures would make the project too ambitious for one semester.
- **Live-workbook modification and monitoring are excluded.** The system receives uploaded copies, so it cannot change or detect changes in an external live workbook. A changed workbook must be uploaded as a new version.
- **Automatic publication is excluded.** The system produces a downloadable remediated copy, but the user remains responsible for reviewing and publishing it. This avoids accidentally replacing public content with an unapproved file.
- **Macro execution is excluded.** Macro-enabled workbooks are not accepted because safely executing uploaded code would create substantial security and testing requirements unrelated to the project's accessibility focus.
- **Automatic legal determinations are excluded.** The system will not decide that a workbook is legally compliant, determine whether an exception under **28 C.F.R. § 35.201** applies, or decide whether a failure has minimal impact under **§ 35.205**. These determinations depend on organizational and legal facts outside the workbook.
- **Guaranteed remediation of every WCAG issue is excluded.** Some barriers depend on human meaning or cannot be reliably evaluated through automated inspection. The system instead identifies those issues for user input or manual review.
- **Real-time collaboration is not applicable.** The hub is an auditing and remediation system, not a replacement for Microsoft Excel or a multi-user spreadsheet editor.
- **Hardware and safety-critical requirements are not applicable.** The system processes electronic documents and does not control physical equipment or perform functions where a processing delay could cause physical harm.
- **Enterprise-level availability is excluded.** Multi-region hosting, formal uptime guarantees, and disaster-recovery infrastructure are disproportionate to a ten-week prototype, although controlled failures and protection of uploaded workbooks remain required.


Workbook Intake and Audit State Machine
Clarifies REQ-01 through REQ-05, REQ-11, REQ-13, REQ-14, and REQ-15
Awaiting Upload
Authenticated upload
Processing
Valid .xlsx
Auditing
No findings
Audit Complete
Human judgment needed
Supported correction found
Review Required
Invalid or unsupported
Rejected
Timeout or error
Remediation Available
Processing Failed
Rejected uploads create no workbook or audit record. Processing failures preserve the uploaded source.
The result determines whether the workflow ends, requires review, or continues to remediation.


Human Review and Remediation State Machine
Clarifies REQ-06 through REQ-11 and REQ-13
Review Required
User supplies meaning
Remediation Available
User approves changes
Remediating
Checks pass
Remediated
Correction declined
Input not supplied; remain in review
Generation or preservation fails
Processing Failed
An audit report remains available for Audit Complete, Review Required, and Remediated states.
The uploaded source is never changed. Remediation creates a separate workbook version.
Only approved changes may appear in the output. Partial failed outputs are not downloadable.


Version, Scheduled Audit, and Reminder State Machine
Clarifies REQ-10, REQ-11, REQ-12, REQ-13, and REQ-14
Failure: log and retain schedule
Stored Version
Enable 1 to 90 day schedule
Audit Scheduled
Scheduled date arrives
Scheduled Audit Processing
Not due: stay scheduled
Success: save new audit record
Changed workbook uploaded
Delivery result recorded
Reminder date arrives
Authorized delete from any stored state
New Version Processing
Notification Pending
Deleted
Create new version
Continue in Model 1
A scheduled audit evaluates only the stored version. External changes require a new upload.
Re-audits create new records. Reminders and delivery failures do not change audit status.

3.3

The workbook intake and audit model defines the initial flow from upload and validation through auditing and classification of the result. The human review and remediation model continues the flow when an audit requires user input or identifies a supported correction, while the version, scheduled audit, and reminder model describes what happens after a workbook version is stored. The models connect when an audit enters Review Required or Remediation Available, when a changed upload begins with the intake flow as a new version, or when a scheduled audit returns a stored version to the auditing process. All implementations must assume that source versions and prior audit records are immutable; failures never produce successful outputs, and scheduled audits evaluate only the copy stored in the hub. Both teams must use the same statuses, accessibility rules, supported remediations, processing limits, and human-review rules, while unresolved decisions such as retry behavior and operational-log retention will be recorded in Section 9.

4.1

AC-01 — Supported workbook acceptance

Requirements evaluated: REQ-01
Setup/input: Prepare valid .xlsx workbooks at or below the supported limits of 25 MB, 50 worksheets, and 250,000 nonempty cells.
Action/check: Upload each workbook using an authenticated account.
Expected result: The system accepts each workbook, creates an audit record, and begins processing.
Method and pass/fail rule: Automated integration test. Pass if every workbook within the limits is accepted and assigned a unique version identifier; fail if any valid workbook is rejected or no audit is created.

AC-02 — Invalid and boundary input rejection

Requirements evaluated: REQ-01, REQ-02
Setup/input: Prepare .xls, .xlsm, CSV, password-protected, encrypted, corrupted, and non-spreadsheet files. Also prepare valid .xlsx files exceeding one supported limit at a time.
Action/check: Attempt to upload each file.
Expected result: The system rejects each file, identifies the reason, and does not begin an audit.
Method and pass/fail rule: Automated negative and boundary testing. Pass if all unsupported

files are rejected with an applicable error and no audit record; fail if an unsupported file is processed or a misleading error is shown.

AC-03 — Original workbook protection

Requirements evaluated: REQ-03, REQ-08
Setup/input: Record the cryptographic hash of a valid workbook before uploading it.
Action/check: Upload, audit, remediate, and download the resulting workbook. Then retrieve or compare the stored source version.
Expected result: The source workbook's hash remains unchanged, and the remediated workbook is stored and downloaded as a separate version.
Method and pass/fail rule: Automated hash comparison and version-record inspection. Pass if the source hash is unchanged and the output has a different version identifier; fail if the source is overwritten or modified.

AC-04 — Accessibility-rule detection

Requirements evaluated: REQ-04
Setup/input: Use shared test workbooks containing known examples of each required issue: missing alternative text, table-header problems, merged cells in populated regions, generic worksheet names, ambiguous hyperlinks, and potential color or contrast problems. Include one workbook without any seeded issues.
Action/check: Audit each test workbook.
Expected result: The system detects every seeded issue covered by the shared rules and does not report those issues in the workbook without them.
Method and pass/fail rule: Automated comparison against the expected-results file defined in Section 5. Pass if all seeded issues are detected at the expected locations and the clean workbook has no false findings for those test cases.

AC-05 — Actionable finding information

Requirements evaluated: REQ-05
Setup/input: Use a workbook containing at least one issue from every required issue category.
Action/check: Inspect the generated findings.
Expected result: Every finding includes the worksheet, affected cell range or object, issue explanation, recommended response, and classification as automatically remediable, requiring user input, or requiring manual review.
Method and pass/fail rule: Automated schema validation followed by human inspection of the explanations. Pass if every finding includes all required fields and the explanations correctly describe the seeded issues.

AC-06 — Meaning-dependent issue handling

Requirements evaluated: REQ-06
Setup/input: Upload a workbook containing an image without alternative text, a generic worksheet name, and information that may be communicated only through color.
Action/check: Complete the audit without supplying additional meaning.
Expected result: The system requests user input or flags each issue for manual review. It does not invent alternative text, rename the worksheet, or decide the meaning of the colors automatically.
Method and pass/fail rule: Automated check of generated changes plus human inspection. Pass if no semantic content is generated without user input and all meaning-dependent issues are presented for review.

AC-07 — User-authorized remediation

Requirements evaluated: REQ-07
Setup/input: Audit a workbook with at least two supported corrections. Approve one correction and decline the other.
Action/check: Generate the remediated workbook.
Expected result: The generated copy contains the approved correction but does not contain the declined correction.
Method and pass/fail rule: Automated comparison of the source and output workbooks. Pass if only the approved change is present; fail if the declined change is applied or the approved change is absent.

AC-08 — Separate downloadable results

Requirements evaluated: REQ-08, REQ-10
Setup/input: Successfully audit and remediate a supported workbook.
Action/check: Request the available downloads.
Expected result: The system provides a separate .xlsx output and an audit report associated with the correct source and output versions.
Method and pass/fail rule: Automated download and metadata test. Pass if both files can be downloaded, have the expected file types, and reference the correct version identifiers.

AC-09 — Workbook-content preservation

Requirements evaluated: REQ-09
Setup/input: Use a workbook containing formulas, values, formatting, named ranges, worksheet references, charts, and other supported features. Approve one known accessibility

correction.
Action/check: Generate the remediated copy, compare it with the source, and open it in Microsoft Excel.
Expected result: The approved correction is present, all unrelated supported content is unchanged, and Excel opens the output without displaying a corruption or repair warning.
Method and pass/fail rule: Automated workbook comparison plus human inspection in Excel. Pass if there are zero unexplained differences and no repair warning.

AC-10 — Complete audit report

Requirements evaluated: REQ-10
Setup/input: Complete an audit containing detected issues, one approved correction, one unresolved issue, and one processing limitation.
Action/check: Generate and inspect the audit report.
Expected result: The report contains the workbook version, audit date, ruleset version, checks performed, findings, approved changes, unresolved issues, manual-review items, and processing limitations.
Method and pass/fail rule: Automated report-schema validation and human inspection. Pass if every required section is present and its contents match the audit record.

AC-11 — Version and status tracking

Requirements evaluated: REQ-11
Setup/input: Upload an original workbook, upload a changed file with the same filename, remediate the second version, and re-audit it.
Action/check: Inspect the version history, audit records, and statuses.
Expected result: Each upload, remediation, and re-audit creates the required separate version or audit record while preserving prior history. Statuses follow the state models and never use “ADA Compliant.”
Method and pass/fail rule: Automated database or API inspection. Pass if identifiers are unique, previous records remain unchanged, and every status is from the defined status set.

AC-12 — Scheduled re-auditing

Requirements evaluated: REQ-12
Setup/input: Store a workbook and configure audit intervals of 1 day and 90 days. Also attempt invalid intervals of 0 and 91 days. Use a simulated clock.
Action/check: Advance the clock to each scheduled date.
Expected result: Valid intervals are accepted, invalid intervals are rejected, and each due audit evaluates the stored version using the current ruleset and creates a new audit record.

Method and pass/fail rule: Automated scheduler test. Pass if audits run only at valid configured intervals and do not overwrite previous records.

AC-13 — Deadline reminders and external-version limitation

Requirements evaluated: REQ-12
Setup/input: Assign a compliance deadline to one workbook with unresolved findings and another with a completed audit and no unresolved findings. Use a simulated clock to reach 90, 30, and 7 days before the deadline.
Action/check: Execute the reminder scheduler and inspect the audit display.
Expected result: Default reminders are requested for the unresolved workbook but not the completed workbook. The display identifies the stored version and states that changes to an external workbook cannot be detected.
Method and pass/fail rule: Automated scheduler and notification-service test plus interface inspection. Pass if reminders are triggered at all three required times only when applicable and the external-change limitation is visible.

AC-14 — Controlled processing failures

Requirements evaluated: REQ-13
Setup/input: Simulate failures during auditing, remediation, report generation, file storage, and scheduled auditing.
Action/check: Run the affected operation for each simulated failure.
Expected result: The system preserves the source, records and displays the error, assigns the appropriate failure status, and does not expose a partial file as successful.
Method and pass/fail rule: Automated fault-injection test. Pass if every failure is recorded and controlled, no source file changes, and no partial output is labeled successful.

AC-15 — User authorization and isolation

Requirements evaluated: REQ-14
Setup/input: Create two standard user accounts and one authorized reviewer account. Upload a workbook through the first standard account.
Action/check: Attempt to access the workbook and report through each account and through an unauthenticated request.
Expected result: The owner and explicitly authorized reviewer can access the records. The second standard user and unauthenticated request are denied.
Method and pass/fail rule: Automated authorization test. Pass if every authorized request succeeds and every unauthorized request returns a denial without revealing workbook content or metadata.

AC-16 — Transport, credential, and deletion security

Requirements evaluated: REQ-14
Setup/input: Deploy the application with a test account and a stored workbook containing generated outputs.
Action/check: Attempt an HTTP connection, inspect password storage, and delete the workbook as its owner.
Expected result: HTTP redirects to HTTPS or is rejected, passwords are not stored in plaintext, and deletion removes workbook files and outputs while canceling scheduled audits.
Method and pass/fail rule: Automated deployment and storage inspection with manual configuration review. Pass only if all three security checks succeed.

AC-17 — Accessible hub interface

Requirements evaluated: REQ-15
Setup/input: Open the upload, findings, remediation, scheduling, and download interfaces using a supported browser, keyboard, and automated accessibility scanner.
Action/check: Complete the primary workflow without a mouse and scan each page.
Expected result: All controls are keyboard-accessible, focus is visible, inputs have programmatic labels, errors are provided as text, and contrast is at least **4.5:1 for normal text** and **3:1 for large text**.
Method and pass/fail rule: Automated accessibility scanning plus manual keyboard inspection. Pass if the workflow can be completed by keyboard, no required control lacks a label or visible focus, and all tested text satisfies the contrast threshold.

AC-18 — Performance and repeatability

Requirements evaluated: REQ-15
Setup/input: Use a valid test workbook no larger than 25 MB with no more than 250,000 nonempty cells in the documented test environment.
Action/check: Audit the workbook five times using the same ruleset version and configuration.
Expected result: Each audit completes within 60 seconds or returns a controlled failure, and all successful runs produce identical issue types and locations.
Method and pass/fail rule: Automated performance and result-comparison test. Pass if all five runs finish within 60 seconds without an uncontrolled failure and the successful findings are identical.

4.2
Representative test data will come from pod-created .xlsx fixtures and de-identified public or institutional-style spreadsheets, including accessible workbooks, files with intentionally seeded accessibility issues, unsupported formats, corrupted files, and workbooks at the documented size and worksheet boundaries. Trustworthy, expected outputs will be established by recording each seeded issue and expected location in a shared results file, then manually confirming them with Microsoft Excel, its Accessibility Checker, and review by at least two pod members. Scheduled audits and notifications will be tested with a controlled clock and mock notification service, so tests do not depend on real dates or external-service availability. The required acceptance tests will not depend on generative AI; if either team experiments with AI-assisted descriptions, those results will be evaluated separately using a fixed model version, prompt, and stored responses.

4.3
Another team can evaluate AC-01 through AC-08, AC-10 through AC-15, and the performance and repeatability portion of AC-18 through the common web-service interface by submitting shared test workbooks and comparing HTTP responses, statuses, findings, reports, version records, and downloadable outputs. AC-09 requires comparing the downloaded source and remediated workbooks and opening the result in Microsoft Excel, while AC-16 requires deployment and security evidence and AC-17 requires automated accessibility scanning and manual keyboard inspection of the user-facing website. These evaluations use only externally observable behavior and supplied evidence, so evaluators do not need access to either team's source code or internal design.

Q5.1 Interface Contract

Every team shall expose the following HTTP interface at /api/v1. Automated tests shall be able to run against either implementation by changing only the service's base URL. All protected operations require a bearer token representing a test user or reviewer. Requests and responses use JSON except for workbook uploads and downloads.

COMMON CONVENTIONS

- Timestamps shall use UTC RFC 3339 format.
- Workbook, version, audit, finding, remediation, and output identifiers shall be unique strings.
- Operations that begin asynchronous processing shall return 202 Accepted.
- Tests shall poll the appropriate status endpoint until processing reaches a terminal state.
- Terminal states are AUDIT_COMPLETE, REVIEW_REQUIRED, REMEDIATION_AVAILABLE, REMEDIATED, and PROCESSING_FAILED.

- Findings shall be ordered by rule identifier, worksheet, and target location.
- Auditing the same workbook with the same ruleset shall produce the same finding identifiers, categories, and locations.

Common error response:

```
{
  "code": "UNSUPPORTED_FILE_TYPE",
  "message": "Only .xlsx workbooks are supported.",
  "requestId": "request-123",
  "details": {}
}
```

Common error status codes:

- 400 Bad Request: Missing or malformed request data
- 401 Unauthorized: Missing or invalid authentication
- 403 Forbidden: User lacks access to the resource
- 404 Not Found: Resource does not exist
- 409 Conflict: Operation is invalid for the resource’s current state
- 413 Content Too Large: Workbook exceeds the supported size
- 415 Unsupported Media Type: File is not an .xlsx workbook
- 422 Unprocessable Content: File is corrupted, encrypted, password-protected, or otherwise unreadable
- 500 Internal Server Error: Controlled internal failure
- 503 Service Unavailable: Service is temporarily unavailable

OPERATION 1: CHECK SERVICE AVAILABILITY

Purpose:
Allows the cross-team test runner to verify that a deployed implementation is reachable.

Related requirements:
REQ-15

Method and path:
GET /api/v1/health

Successful response:

200 OK

```
{
  "status": "AVAILABLE",
  "apiVersion": "1.0",
  "rulesetVersion": "1.0"
}
```

Error responses:
- 503 Service Unavailable: The service cannot currently accept requests.

State change:
None.

OPERATION 2: UPLOAD AN INITIAL WORKBOOK

Purpose:
Creates a workbook, immutable source version, and initial accessibility audit.

Related requirements:
REQ-01, REQ-02, REQ-03, REQ-04, REQ-11, REQ-13

Method and path:
POST /api/v1/workbooks

Request format:
Content-Type: multipart/form-data

file: Binary .xlsx workbook

Successful response:
202 Accepted

```
{
  "workbookId": "workbook-001",
  "versionId": "version-001",
  "auditId": "audit-001",
  "status": "PROCESSING"
}
```

Error responses:
- 400: File field is missing.
- 401: User is not authenticated.
- 413: File exceeds the supported size.
- 415: File is not an .xlsx workbook.
- 422: Workbook is corrupted, encrypted, password-protected, macro-enabled, or exceeds another documented processing limit.
- 500: Upload or storage fails.

State change:
A successful request creates an immutable source version and an audit in the PROCESSING state. A rejected upload creates no workbook, version, or audit record.

OPERATION 3: UPLOAD A NEW WORKBOOK VERSION

Purpose:
Adds a changed workbook to an existing workbook history without replacing prior versions.

Related requirements:
REQ-03, REQ-10, REQ-11, REQ-13, REQ-14

Method and path:
POST /api/v1/workbooks/{workbookId}/versions

Request format:
Content-Type: multipart/form-data

file: Binary .xlsx workbook

Successful response:
202 Accepted

```
{
  "workbookId": "workbook-001",
  "versionId": "version-002",
  "auditId": "audit-002",
  "status": "PROCESSING"
}
```

Error responses:
- 400: File field is missing.
- 401: User is not authenticated.
- 403: User cannot add a version to this workbook.
- 404: Workbook does not exist.
- 413: File exceeds the supported size.
- 415: File is not an .xlsx workbook.
- 422: File is invalid or otherwise unsupported.
- 500: Storage or audit initialization fails.

State change:
Creates a new immutable source version and audit. Previous versions and audit records remain unchanged.

OPERATION 4: RETRIEVE A WORKBOOK VERSION

Purpose:
Returns the externally observable state and metadata of one workbook version.

Related requirements:
REQ-03, REQ-11, REQ-14

Method and path:
GET /api/v1/workbooks/{workbookId}/versions/{versionId}

Successful response:
200 OK

```
{
  "workbookId": "workbook-001",
  "versionId": "version-001",
  "fileName": "budget.xlsx",
  "sha256": "9ae056...",
  "createdAt": "2026-09-20T18:00:00Z",
  "status": "REMEDIATION_AVAILABLE",
  "latestAuditId": "audit-001",
  "sourceVersionId": null
}
```

Error responses:
- 401: User is not authenticated.
- 403: User cannot access the workbook.
- 404: Workbook or version does not exist.

State change:
None.

OPERATION 5: RETRIEVE AN AUDIT AND ITS FINDINGS

Purpose:
Allows tests to poll audit status and inspect findings and report data.

Related requirements:
REQ-04, REQ-05, REQ-06, REQ-10, REQ-11, REQ-13

Method and path:
GET /api/v1/workbooks/{workbookId}/versions/{versionId}/audits/{auditId}

Successful response:
200 OK


```
{
  "auditId": "audit-001",
  "workbookId": "workbook-001",
  "versionId": "version-001",
  "rulesetVersion": "1.0",
  "status": "REMEDIATION_AVAILABLE",
  "startedAt": "2026-09-20T18:00:01Z",
  "completedAt": "2026-09-20T18:00:10Z",
  "findings": [
    {
      "findingId": "ALT_TEXT-Sheet1-Chart1",
      "ruleId": "ALT_TEXT",
      "category": "NON_TEXT_CONTENT",
      "worksheet": "Sheet1",
      "target": "Chart1",
      "classification": "REQUIRES_USER_INPUT",
```

```
    "message": "Chart1 does not contain alternative text.",
    "recommendation": "Provide a concise description of the chart."
   }
  ],
  "limitations": [],
  "error": null
}
```


Error responses:
- 401: User is not authenticated.
- 403: User cannot access the audit.
- 404: Workbook, version, or audit does not exist.

State change:
None. While processing, the response uses the PROCESSING status and may contain an empty findings list.

OPERATION 6: TRIGGER A NEW AUDIT

Purpose:
Re-audits an existing stored version without requiring another workbook upload.

Related requirements:
REQ-04, REQ-10, REQ-11, REQ-12, REQ-13

Method and path:
POST /api/v1/workbooks/{workbookId}/versions/{versionId}/audits

Optional request:


```
{
  "reason": "MANUAL_REAUDIT"
}
```


Successful response:
202 Accepted


```
{
  "auditId": "audit-003",
```

```
  "versionId": "version-001",
  "rulesetVersion": "1.1",
  "status": "PROCESSING"
}
```


Error responses:
- 400: Request data is malformed.
- 401: User is not authenticated.
- 403: User cannot audit this workbook.
- 404: Workbook or version does not exist.
- 409: Another audit is already processing for this version.
- 500: Audit cannot be started.

State change:
Creates a new audit record without changing or deleting previous audit records.

OPERATION 7: SUBMIT REMEDIATION DECISIONS

Purpose:
Supplies human input and identifies which supported corrections may be applied.

Related requirements:
REQ-06, REQ-07, REQ-08, REQ-09, REQ-10, REQ-13

Method and path:
POST /api/v1/workbooks/{workbookId}/versions/{versionId}/remediations

Request format:
Content-Type: application/json


```
{
  "auditId": "audit-001",
  "decisions": [
    {
      "findingId": "ALT_TEXT-Sheet1-Chart1",
      "action": "APPLY",
      "value": "Bar chart comparing monthly expenses."
    },
    {
```

```
    "findingId": "LINK_TEXT-Sheet1-A15",
    "action": "DECLINE"
   }
  ]
}
```

Allowed actions:

- APPLY: Apply the supported correction. A value is required when the issue requires user-provided meaning.
- DECLINE: Leave the issue unresolved.
- MANUAL_REVIEW: Record that the issue must be reviewed outside the automated remediation process.

Successful response:
202 Accepted

```
{
 "remediationId": "remediation-001",
 "sourceVersionId": "version-001",
 "status": "REMEDIATING"
}
```

Error responses:
- 400: A decision, action, or required value is missing.
- 401: User is not authenticated.
- 403: User cannot remediate this workbook.
- 404: Workbook, version, audit, or finding does not exist.
- 409: Audit is not in a state that permits remediation.
- 422: Provided value is invalid for the selected correction.
- 500: Remediation cannot be started.

State change:
The source version remains unchanged. Successful processing creates a new remediated version. A failure creates no successful output.

OPERATION 8: RETRIEVE REMEDIATION STATUS

Purpose:

Allows tests to determine whether remediation succeeded or failed.

Related requirements:
REQ-08, REQ-09, REQ-10, REQ-11, REQ-13

Method and path:
GET /api/v1/remediations/{remediationId}

Successful response:
200 OK


```
{
  "remediationId": "remediation-001",
  "status": "REMEDIATED",
  "sourceVersionId": "version-001",
  "outputVersionId": "version-002",
  "outputId": "output-001",
  "auditId": "audit-004",
  "error": null
}
```


Error responses:
- 401: User is not authenticated.
- 403: User cannot access the remediation.
- 404: Remediation does not exist.

State change:
None. While processing, the status is REMEDIATING. If generation or preservation checks fail, the status is PROCESSING_FAILED, outputId is null, and the error field explains the failure.

OPERATION 9: DOWNLOAD THE REMEDIATED WORKBOOK

Purpose:
Returns the generated .xlsx file for independent workbook comparison and Excel inspection.

Related requirements:
REQ-08, REQ-09, REQ-13, REQ-14

Method and path:

GET /api/v1/outputs/{outputId}/workbook

Successful response:
200 OK

Content-Type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
Content-Disposition: attachment; filename="budget-remediated.xlsx"

Response body:
Binary .xlsx content

Error responses:
- 401: User is not authenticated.
- 403: User cannot access the output.
- 404: Output does not exist.
- 409: Output is not ready or remediation failed.

State change:
None. Downloading the workbook does not change its status.

OPERATION 10: RETRIEVE VERSION AND AUDIT HISTORY

Purpose:
Allows tests to verify that earlier versions and audits are preserved.

Related requirements:
REQ-10, REQ-11, REQ-14

Method and path:
GET /api/v1/workbooks/{workbookId}/history

Successful response:
200 OK

```
{
  "workbookId": "workbook-001",
  "versions": [
    {
      "versionId": "version-001",
```

```
    "sourceVersionId": null,
    "status": "REMEDIATION_AVAILABLE",
    "createdAt": "2026-09-20T18:00:00Z"
   },
   {
    "versionId": "version-002",
    "sourceVersionId": "version-001",
    "status": "REMEDIATED",
    "createdAt": "2026-09-20T18:10:00Z"
   }
  ],
  "audits": [
   {
    "auditId": "audit-001",
    "versionId": "version-001",
    "rulesetVersion": "1.0",
    "status": "REMEDIATION_AVAILABLE"
   },
   {
    "auditId": "audit-004",
    "versionId": "version-002",
    "rulesetVersion": "1.0",
    "status": "AUDIT_COMPLETE"
   }
  ]
}
```


Error responses:
- 401: User is not authenticated.
- 403: User cannot access the workbook history.
- 404: Workbook does not exist.

State change:
None.

OPERATION 11: CONFIGURE PERIODIC AUDITING AND REMINDERS

Purpose:
Enables or disables scheduled auditing and records the applicable compliance deadline.

Related requirements:
REQ-12, REQ-14

Method and path:
PUT /api/v1/workbooks/{workbookId}/schedule

Request format:
Content-Type: application/json

```
{
  "enabled": true,
  "intervalDays": 30,
  "complianceDeadline": "2027-04-26"
}
```

Successful response:
200 OK

```
{
  "workbookId": "workbook-001",
  "enabled": true,
  "intervalDays": 30,
  "complianceDeadline": "2027-04-26",
  "nextAuditAt": "2026-10-20T18:00:00Z",
  "reminderDays": [90, 30, 7]
}
```

Error responses:
- 400: Interval is outside 1–90 days or the date is invalid.
- 401: User is not authenticated.
- 403: User cannot modify the workbook schedule.
- 404: Workbook does not exist.

State change:
Enabling the schedule creates the next audit date. Disabling the schedule cancels future scheduled audits but preserves prior audit records.

OPERATION 12: RETRIEVE NOTIFICATION RECORDS

Purpose:
Allows automated tests to verify that reminders were requested without depending on actual email delivery.

Related requirements:
REQ-12, REQ-13, REQ-14

Method and path:
GET /api/v1/workbooks/{workbookId}/notifications

Successful response:
200 OK


```
{
  "notifications": [
    {
      "notificationId": "notification-001",
      "type": "COMPLIANCE_DEADLINE_REMINDER",
      "workbookId": "workbook-001",
      "versionId": "version-002",
      "scheduledFor": "2027-01-26T00:00:00Z",
      "status": "DELIVERED",
      "message": "This workbook has unresolved accessibility findings."
    }
  ]
}
```


Error responses:
- 401: User is not authenticated.
- 403: User cannot access the notifications.
- 404: Workbook does not exist.

State change:
None. Notification success or failure does not change the workbook’s audit status.

OPERATION 13: ADVANCE THE TEST CLOCK

Purpose:

Supports repeatable tests of scheduled audits and reminders without waiting for real dates. This operation is available only in the designated test deployment.

Related requirements:
REQ-12, REQ-15

Method and path:
POST /api/v1/test/clock

Request format:
Content-Type: application/json

```
{
  "currentTime": "2027-01-26T00:00:00Z"
}
```

Successful response:
200 OK

```
{
  "currentTime": "2027-01-26T00:00:00Z",
  "triggeredAudits": [],
  "triggeredNotifications": ["notification-001"]
}
```

Error responses:
- 400: Timestamp is invalid.
- 401: User is not authenticated.
- 403: User does not have the shared test-runner credential.
- 404: The test-clock operation is disabled outside the test environment.

State change:
Updates the test environment’s simulated time and triggers audits or reminders due at or before that time.

OPERATION 14: DELETE A WORKBOOK

Purpose:
Deletes stored workbook content and cancels future scheduled operations.

Related requirements:
REQ-12, REQ-14

Method and path:
DELETE /api/v1/workbooks/{workbookId}

Successful response:
204 No Content

Error responses:
- 401: User is not authenticated.
- 403: User is not authorized to delete the workbook.
- 404: Workbook does not exist.
- 500: Deletion cannot be completed.

State change:
Removes the workbook’s source files, remediated outputs, and downloadable reports and cancels scheduled audits. Only permitted non-content operational logs may remain.

STATE INTERPRETATION RULES

- PROCESSING: The audit has started, but results are not ready.
- AUDIT_COMPLETE: The supported audit completed with no unresolved automated findings. This status does not establish legal compliance.
- REVIEW_REQUIRED: At least one finding requires human judgment or remains unresolved.
- REMEDIATION_AVAILABLE: At least one supported correction can be submitted.
- REMEDIATING: Approved corrections are being applied to a new workbook copy.
- REMEDIATED: A separate output passed generation and preservation checks.
- PROCESSING_FAILED: Processing ended in a controlled failure, and no output may be treated as successful.
- REJECTED: The upload was invalid and produced no stored workbook, version, or audit resource.

5.2
Q5.2 Data Schemas and Examples

The HTTP methods, paths, purposes, state changes, and status codes are defined in Q5.1. This section defines the shared data types used by those operations.

COMMON DATA TYPES

Identifier
- Type: String
- Required format: 1–64 characters matching [A-Za-z0-9_-]+

Timestamp
- Type: String
- Required format: RFC 3339 UTC
- Example: 2026-09-20T18:00:00Z

Date
- Type: String
- Required format: YYYY-MM-DD
- Example: 2027-04-26

SHA-256
- Type: String
- Required format: 64 lowercase hexadecimal characters

File size
- Type: Integer
- Unit: Bytes
- Maximum: 25,000,000 bytes

Binary workbook
- Type: Binary file
- Required format: Valid Office Open XML .xlsx workbook

WORKBOOK UPLOAD SCHEMA

Used by Operations 2 and 3 in Q5.1.

file
- Type: Binary workbook
- Status: Required

- Constraints: Must be a valid .xlsx workbook no larger than 25,000,000 bytes, containing no more than 50 worksheets and 250,000 nonempty cells.

WORKBOOK UPLOAD RESPONSE SCHEMA

workbookId
- Type: Identifier
- Status: Required
- Description: Identifies the workbook and its version history.

versionId
- Type: Identifier
- Status: Required
- Description: Identifies the immutable uploaded version.

auditId
- Type: Identifier
- Status: Required
- Description: Identifies the initial accessibility audit.

status
- Type: String
- Status: Required
- Allowed value for an accepted upload: PROCESSING

WORKBOOK VERSION SCHEMA

Used by Operations 4 and 10 in Q5.1.

workbookId
- Type: Identifier
- Status: Required
- Description: Identifies the parent workbook.

versionId
- Type: Identifier
- Status: Required
- Description: Identifies the exact workbook version.

fileName
- Type: String
- Status: Required
- Constraints: 1–255 characters.

sha256
- Type: SHA-256 string
- Status: Required
- Description: Hash of the exact stored file.

createdAt
- Type: Timestamp
- Status: Required
- Description: Time at which the version was created.

status
- Type: String
- Status: Required
- Allowed values:
  - PROCESSING
  - AUDIT_COMPLETE
  - REVIEW_REQUIRED
  - REMEDIATION_AVAILABLE
  - REMEDIATING
  - REMEDIATED
  - PROCESSING_FAILED

latestAuditId
- Type: Identifier
- Status: Optional
- Description: Identifies the latest audit.
- Null when no audit exists.

sourceVersionId
- Type: Identifier
- Status: Optional
- Description: Identifies the source of a remediated version.
- Null for a directly uploaded version.

REJECTED is not a stored version status because a rejected upload does not create a version.

ACCESSIBILITY FINDING SCHEMA

Used in the audit response from Operation 5.

findingId
- Type: Identifier
- Status: Required
- Description: Stable identifier for the same rule, ruleset, workbook, and location.

ruleId
- Type: String
- Status: Required
- Description: Rule identifier from the shared accessibility-rule catalog.

category
- Type: String
- Status: Required
- Allowed values:
  - NON_TEXT_CONTENT
  - TABLE_STRUCTURE
  - MERGED_CELLS
  - WORKSHEET_NAME
  - LINK_TEXT
  - COLOR_OR_CONTRAST

worksheet
- Type: String
- Status: Required
- Description: Name of the worksheet containing the finding.

target
- Type: String
- Status: Required
- Description: Cell, range, table, chart, image, hyperlink, or worksheet identifier.

classification
- Type: String

- Status: Required
- Allowed values:
  - AUTOMATICALLY_REMEDIABLE
  - REQUIRES_USER_INPUT
  - MANUAL_REVIEW

message
- Type: String
- Status: Required
- Constraints: 1–500 characters.
- Description: Plain-language explanation of the finding.

recommendation
- Type: String
- Status: Required
- Constraints: 1–500 characters.
- Description: Recommended response to the finding.

AUDIT RECORD SCHEMA

Used by Operations 5, 6, and 10 in Q5.1.

auditId
- Type: Identifier
- Status: Required

workbookId
- Type: Identifier
- Status: Required

versionId
- Type: Identifier
- Status: Required
- Description: Exact workbook version evaluated by the audit.

rulesetVersion
- Type: String
- Status: Required
- Format: Semantic version, such as 1.0

status
- Type: String
- Status: Required
- Allowed values:
  - PROCESSING
  - AUDIT_COMPLETE
  - REVIEW_REQUIRED
  - REMEDIATION_AVAILABLE
  - PROCESSING_FAILED

startedAt
- Type: Timestamp
- Status: Required

completedAt
- Type: Timestamp
- Status: Optional
- Null while the audit is processing.

findings
- Type: Array of accessibility findings
- Status: Required
- Empty while processing or when no supported issues are detected.

limitations
- Type: Array of strings
- Status: Required
- Empty when no processing limitations apply.

error
- Type: Error object
- Status: Optional
- Null unless processing fails.

REMEDIATION DECISION REQUEST SCHEMA

Used by Operation 7 in Q5.1.

auditId
- Type: Identifier
- Status: Required
- Description: Identifies the audit containing the findings.

decisions
- Type: Array
- Status: Required
- Constraint: Must contain at least one decision.

decisions[].findingId
- Type: Identifier
- Status: Required
- Constraint: Must identify a finding in the specified audit.

decisions[].action
- Type: String
- Status: Required
- Allowed values:
  - APPLY
  - DECLINE
  - MANUAL_REVIEW

decisions[].value
- Type: String
- Status: Conditional
- Constraint: Maximum 1,000 characters.
- Required when APPLY needs user-provided meaning.
- Omitted for DECLINE and MANUAL_REVIEW.

REMEDIATION RESPONSE SCHEMA

Used by Operations 7 and 8 in Q5.1.

remediationId
- Type: Identifier
- Status: Required

status

- Type: String
- Status: Required
- Allowed values:
  - REMEDIATING
  - REMEDIATED
  - PROCESSING_FAILED

sourceVersionId
- Type: Identifier
- Status: Required
- Description: Immutable workbook version being remediated.

outputVersionId
- Type: Identifier
- Status: Optional
- Null until remediation succeeds.

outputId
- Type: Identifier
- Status: Optional
- Null until a downloadable output exists.

auditId
- Type: Identifier
- Status: Optional
- Description: Identifies the audit of the generated output.
- Null until the audit exists.

error
- Type: Error object
- Status: Optional
- Null unless remediation fails.

SCHEDULE REQUEST AND RESPONSE SCHEMA

Used by Operation 11 in Q5.1.

enabled
- Type: Boolean

- Status: Required
- Description: Enables or disables future scheduled audits.

intervalDays
- Type: Integer
- Status: Conditional
- Required when enabled is true.
- Allowed range: 1–90 days, inclusive.

complianceDeadline
- Type: Date
- Status: Optional
- Description: Administrator-provided compliance deadline.

nextAuditAt
- Type: Timestamp
- Status: Response only
- Null when scheduled auditing is disabled.

reminderDays
- Type: Array of integers
- Status: Response only
- Default value: [90, 30, 7]

NOTIFICATION RECORD SCHEMA

Used by Operation 12 in Q5.1.

notificationId
- Type: Identifier
- Status: Required

type
- Type: String
- Status: Required
- Allowed values:
  - COMPLIANCE_DEADLINE_REMINDER
  - AUDIT_REMINDER

workbookId
- Type: Identifier
- Status: Required

versionId
- Type: Identifier
- Status: Required
- Description: Stored version referenced by the notification.

scheduledFor
- Type: Timestamp
- Status: Required

status
- Type: String
- Status: Required
- Allowed values:
  - PENDING
  - DELIVERED
  - FAILED

message
- Type: String
- Status: Required
- Maximum length: 500 characters.

ERROR OBJECT SCHEMA

Used by every error response defined in Q5.1.

code
- Type: String
- Status: Required
- Description: Stable machine-readable error code.

message
- Type: String
- Status: Required
- Description: Human-readable explanation.

requestId
- Type: Identifier
- Status: Required
- Description: Identifies the failed request for tracing.

details
- Type: Object
- Status: Optional
- Description: Additional structured error information.

VALID CORE-OPERATION EXAMPLES

EXAMPLE 1: UPLOAD A WORKBOOK

Request:

POST /api/v1/workbooks
Authorization: Bearer test-user-a
Content-Type: multipart/form-data

file = budget.xlsx

Response:

{
  "workbookId": "workbook-001",
  "versionId": "version-001",
  "auditId": "audit-001",
  "status": "PROCESSING"
}

Expected status:
202 Accepted

EXAMPLE 2: RETRIEVE A COMPLETED AUDIT

Request:

GET /api/v1/workbooks/workbook-001/versions/version-001/audits/audit-001
Authorization: Bearer test-user-a

Response:


```
{
  "auditId": "audit-001",
  "workbookId": "workbook-001",
  "versionId": "version-001",
  "rulesetVersion": "1.0",
  "status": "REMEDIATION_AVAILABLE",
  "startedAt": "2026-09-20T18:00:01Z",
  "completedAt": "2026-09-20T18:00:10Z",
  "findings": [
    {
      "findingId": "ALT_TEXT-Sheet1-Chart1",
      "ruleId": "ALT_TEXT",
      "category": "NON_TEXT_CONTENT",
      "worksheet": "Sheet1",
      "target": "Chart1",
      "classification": "REQUIRES_USER_INPUT",
      "message": "Chart1 does not contain alternative text.",
      "recommendation": "Provide a concise description of the chart."
    }
  ],
  "limitations": [],
  "error": null
}
```


Expected status:
200 OK

EXAMPLE 3: SUBMIT A REMEDIATION DECISION

Request:

POST /api/v1/workbooks/workbook-001/versions/version-001/remediations
Authorization: Bearer test-user-a

Content-Type: application/json


```
{
  "auditId": "audit-001",
  "decisions": [
    {
      "findingId": "ALT_TEXT-Sheet1-Chart1",
      "action": "APPLY",
      "value": "Bar chart comparing monthly expenses."
    }
  ]
}
```


Response:


```
{
  "remediationId": "remediation-001",
  "status": "REMEDIATING",
  "sourceVersionId": "version-001",
  "outputVersionId": null,
  "outputId": null,
  "auditId": null,
  "error": null
}
```


Expected status:
202 Accepted

EXAMPLE 4: RETRIEVE SUCCESSFUL REMEDIATION

Request:

GET /api/v1/remediations/remediation-001
Authorization: Bearer test-user-a

Response:


```
{
  "remediationId": "remediation-001",
```

```
  "status": "REMEDIATED",
  "sourceVersionId": "version-001",
  "outputVersionId": "version-002",
  "outputId": "output-001",
  "auditId": "audit-002",
  "error": null
}
```

Expected status:
200 OK

EXAMPLE 5: DOWNLOAD THE REMEDIATED WORKBOOK

Request:

```
GET /api/v1/outputs/output-001/workbook
Authorization: Bearer test-user-a
```

Response:

```
200 OK
Content-Type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
Content-Disposition: attachment; filename="budget-remediated.xlsx"

[Binary .xlsx content]
```

EXAMPLE 6: CONFIGURE PERIODIC AUDITING

Request:

```
PUT /api/v1/workbooks/workbook-001/schedule
Authorization: Bearer test-user-a
Content-Type: application/json

{
  "enabled": true,
  "intervalDays": 30,
  "complianceDeadline": "2027-04-26"
```

}

Response:


```
{
  "workbookId": "workbook-001",
  "enabled": true,
  "intervalDays": 30,
  "complianceDeadline": "2027-04-26",
  "nextAuditAt": "2026-10-20T18:00:00Z",
  "reminderDays": [90, 30, 7]
}
```


Expected status:
200 OK

INVALID-INPUT EXAMPLES

INVALID EXAMPLE 1: UNSUPPORTED WORKBOOK

Request:

POST /api/v1/workbooks
Authorization: Bearer test-user-a
Content-Type: multipart/form-data

file = budget.xlsm

Response:


```
{
  "code": "MACRO_ENABLED_WORKBOOK_UNSUPPORTED",
  "message": "Macro-enabled workbooks are outside the supported project scope.",
  "requestId": "request-201",
  "details": {
    "fileName": "budget.xlsm"
  }
}
```

Expected status:
422 Unprocessable Content

Expected state:
No workbook, version, or audit record is created.

INVALID EXAMPLE 2: MISSING REMEDIATION VALUE

Request:

POST /api/v1/workbooks/workbook-001/versions/version-001/remediations
Authorization: Bearer test-user-a
Content-Type: application/json


```
{
  "auditId": "audit-001",
  "decisions": [
    {
      "findingId": "ALT_TEXT-Sheet1-Chart1",
      "action": "APPLY"
    }
  ]
}
```


Response:


```
{
  "code": "REMEDIATION_VALUE_REQUIRED",
  "message": "A value is required to apply this remediation.",
  "requestId": "request-202",
  "details": {
    "findingId": "ALT_TEXT-Sheet1-Chart1",
    "field": "value"
  }
}
```


Expected status:
400 Bad Request

Expected state:
The source workbook remains unchanged, and no remediation operation is created.

5.3

The test runner will select an implementation through a configurable base URL, obtain shared bearer tokens for Test User A, Test User B, and the authorized reviewer, and first confirm availability through the health endpoint. Each test run will use uniquely named workbook fixtures and will reset state by deleting workbooks created during the previous run and restoring the test clock to the scenario's starting time. Shared fixtures will include valid, inaccessible, corrupted, unsupported, and boundary-sized workbooks with agreed expected findings; notification behavior will be checked through notification records rather than depending on actual email delivery. Stateful tests must run in workflow order: upload, poll for audit completion, submit remediation, poll for output, and then download or inspect history. However, the tests will depend only on the common HTTP contract, not either team's internal architecture or implementation technology.

Q6.1 Phase 3

In Phase 3, the team will deploy authentication, .xlsx upload and validation, and audits for missing alternative text and generic worksheet names, partially covering REQ-01 through REQ-05, REQ-11, REQ-13, and REQ-14. The team will publish the service URL and test runner and update the interface documentation, system models, fixtures, deployment instructions, and known limitations. This phase will determine whether the selected library can reliably parse workbooks and produce deterministic finding locations. Completion will be demonstrated by uploading valid and invalid files, retrieving the seeded findings, and confirming that auditing does not alter the uploaded workbook's hash. Automated validation will include AC-01, AC-02, the audit-only portion of AC-03, the Phase 3 portions of AC-04 and AC-05, and health, authentication, and upload-failure tests.

Q6.2 Phase 4

In Phase 4, the team will add the remaining shared audit rules, human-review workflow, approved remediation, downloadable workbook copies, audit reports, and version history, completing REQ-03 through REQ-11. The deployed service, test runner, interface schemas, workflow models, remediation rules, fixtures, and known limitations will be updated. This phase will determine whether supported corrections can be applied without altering unrelated formulas, values, formatting, references, or workbook behavior. Completion will be demonstrated by

auditing a seeded workbook, providing human input, approving selected corrections, downloading the output, and confirming that the original workbook remains unchanged while the new version preserves unrelated content. Automated validation will complete AC-03 through AC-11 and rerun AC-01 and AC-02 as regression tests.

Q6.3 Phase 5

In Phase 5, the team will add scheduled audits, deadline reminders, controlled failures, access controls, deletion, interface accessibility, and performance checks, completing REQ-12 through REQ-15. The deployment, test runner, schedule model, security documentation, accessibility evidence, performance results, and known limitations will be updated. This phase will evaluate scheduling reliability, notification behavior, user-data isolation, failure recovery, and performance on supported workbooks. Completion will be demonstrated by advancing the test clock, triggering an audit and reminder, verifying cross-user access denial, simulating a processing failure, and completing the interface workflow by keyboard. Automated validation will implement AC-12 through AC-18 and rerun AC-01 through AC-11 as regression tests.

6.4

This order is feasible because Phase 3 establishes the deployed service, common interface, authentication, workbook parsing, and basic auditing needed by every later capability. It also reduces the highest early technical risk by confirming that both teams can process .xlsx files and produce consistent findings. Phase 4 builds remediation and versioning only after the audit results are stable enough to identify and verify specific changes. Phase 5 then adds scheduling, security, accessibility, and performance validation around the completed primary workflow. If progress falls behind, the pod will reduce the number of supported remediation types, lower the maximum workbook limits, or use only in-application notification records instead of external email delivery while preserving the core upload, audit, review, downloadable-copy, and reporting workflow.

Q10: References

Format each reference as follows: [Number] Author or issuing organization. “Title.” Publication date or version, if available. URL. Accessed: YYYY-MM-DD.

[1] U.S. Department of Justice Civil Rights Division. “Fact Sheet: New Rule on the Accessibility of Web Content and Mobile Apps Provided by State and Local Governments.” April 08, 2024. https://www.ada.gov/resources/2024-03-08-web-rule/. Accessed: 2026-09-27.

[2] World Wide Web Consortium. “Web Content Accessibility Guidelines (WCAG) 2.1.” May 06, 2025. https://www.w3.org/TR/WCAG21/. Accessed: 2026-09-27.

[3] World Wide Web Consortium. “How to Meet WCAG (Quick Reference).”

September 22, 2025. https://www.w3.org/WAI/WCAG22/quickref/?versions=2.1. Accessed: 2026-09-27.